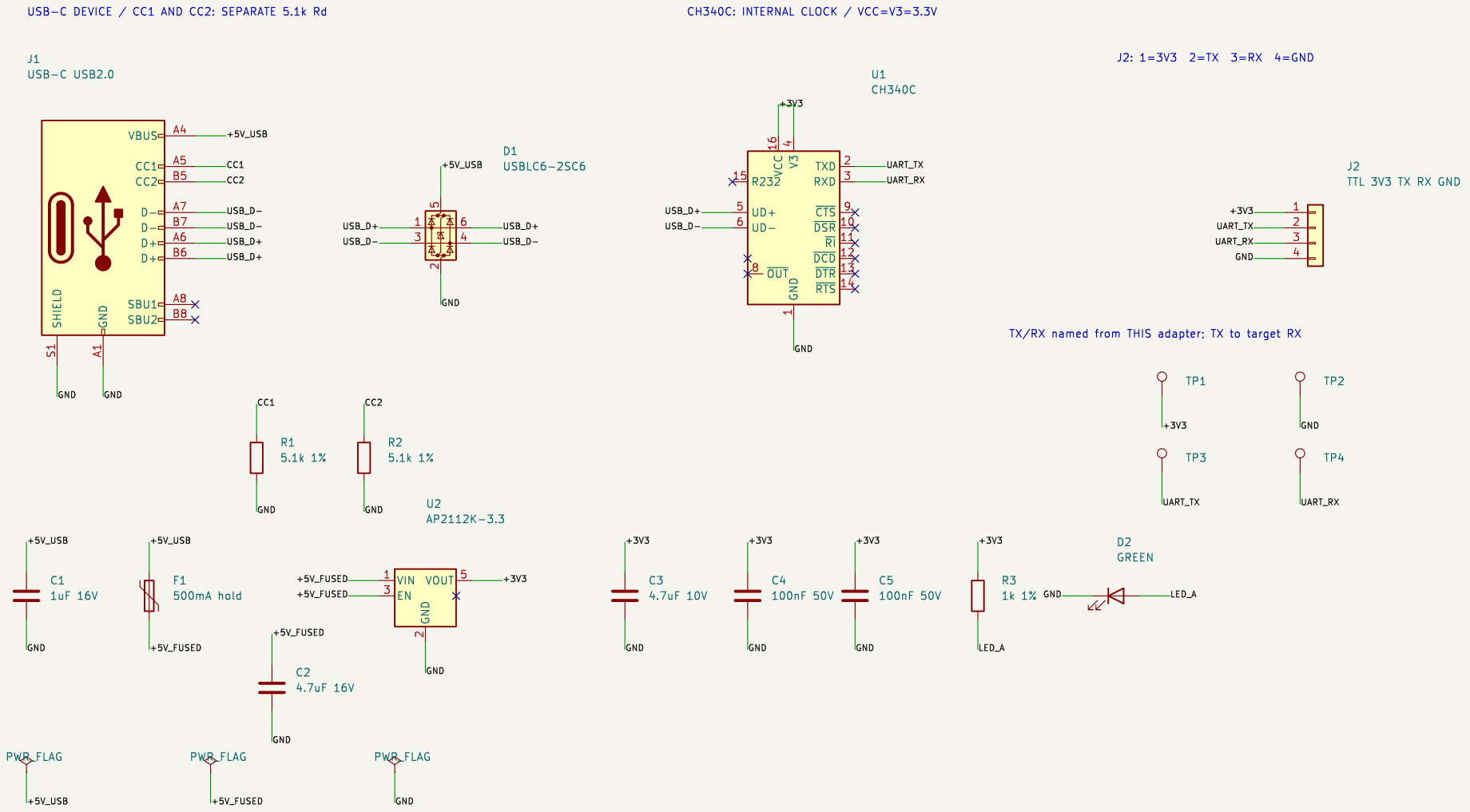

USB-C DEVICE / CC1 AND CC2: SEPARATE 5.1k Rd
J1
USB-C USB2.0
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
S1
A1
A4
A5
B5
A7
B7
A6
B6
A8
B8
+5V_USB
CC1
CC2
USB_D-
USB_D-
USB_D+
USB_D+
GND
D1
USBLC6-2SC6
R1
5.1k 1%
R2
5.1k 1%
U2
AP2112K-3.3
C1
1uF 16V
F1
500mA hold
C2
4.7uF 16V
VIN
VOUT
EN
+5V_FUSED
+3V3
PWR_FLAG
CH340C: INTERNAL CLOCK / VCC=V3=3.3V
U1
CH340C
R232
VCC
V3
UD+
UD-
OUT
TXD
RXD
CTS
DSR
RI
DCD
DTR
RTS
UART_TX
UART_RX
J2: 1=3V3 2=TX 3=RX 4=GND
J2
TTL 3V3 TX RX GND
TX/RX named from THIS adapter; TX to target RX
TP1
TP2
TP3
TP4
C3
4.7uF 10V
C4
100nF 50V
C5
100nF 50V
R3
1k 1%
D2
GREEN
LED_A
USB 5V -> PTC -> LDO 3V3; external output design budget 50 mA
USB D+/D- direct: no series resistors; target 90 ohm differential
Internal clock; unused I/O left open per WCH. Use UART without hardware flow control.
30 x 30 mm / 2 layers / 1.0 mm
Sheet: /
File: ch340-usb-ttl.kicad_sch
Title: CH340C USB-C to 3.3V TTL UART
Size: A4
Date: 2026-11-10
Rev: A
KiCad E.D.A. 9.0.2
Id: 1/1